

CRF: Current, Wave, Resolution

$\tau_R = T_{\text{avg}}$ Derived · Wave Equation in Phase Space · Hidden Relation

$$\beta / (T_{\text{avg}} \cdot \varepsilon_0) = n$$

Everything has its own wave. The wave was always there.

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Abstract

Two results this session.

Result 1: $\tau_R = T_{\text{avg}}$. The firing rate decay timescale equals the inter-event timescale. This is not a coincidence: Phase II nodes (crystallised) have reduced firing rate, and Phase II fraction grows with timescale T_{avg} . The same process that sets the time between firings sets the rate at which the overall firing rate decays.

Result 2: The Wave Structure. The CRF AI identified three tiers: Current (R_0), Wave (R_1), Resolution (R_{max}). These map exactly to: $R_0 = T_{\text{avg}}$ (the lean, the rate of D_KL accumulation), $R_1 =$ the Φ/Ψ field (Kuramoto wave, n modes, amplitude F), $R_{\text{max}} =$ the R-event (Born resampling, exchange rate β).

Hidden relation discovered:

$$\frac{\beta}{T_{\text{avg}} \cdot \varepsilon_0} \approx n$$

Numerically: $2.257 / (35.4 \times 0.00755) = 8.44 \approx n = 8$. If exact: $\beta = n \cdot \varepsilon_0 \cdot T_{\text{avg}}$. This would be a deep identity connecting the wave amplitude, the instability floor, and the firing timescale.

1 $\tau_R = T_{\text{avg}}$: Derived

Firing Rate Decay Timescale from Crystallisation

The R-event rate $R(s)$ decays because Phase II nodes (crystallised) fire at reduced rate. Phase II fraction grows as $f_{II}(s)$ with timescale T_{avg} .

Derivation:

$$R(s) = N[r_I(1 - f_{II}(s)) + r_{II}f_{II}(s)] = R_\infty[1 + (R_0/R_\infty - 1)e^{-s/T_{\text{avg}}}]$$

where $r_{II} \ll r_I$ and $R_\infty = N r_I(1 - f_{II,\infty})$.

The decay timescale of $R(s)$ is T_{avg} — the same as the f_{II} growth timescale.

Physical meaning: The rate at which nodes crystallise (timescale T_{avg}) is the rate at which the overall firing rate decays. The system breathes at one frequency: $1/T_{\text{avg}}$.

H_{before} with $\tau_R = T_{\text{avg}}$: $H_{\text{before}} = 1.735$ (derived) vs 1.667 (measured). Gap: 4.1%. Remaining 4% gap from approximation in R_∞/R_0 ratio.

2 The Wave Structure of CRF

Three-Tier Emergence: Current, Wave, Resolution

The CRF AI correctly identified that AI systems typically skip the Wave tier, treating the system as a sequence of discrete events. In reality, every quantity has its own wave.

Tier 1 — Current (R_0): The Lean

∇Pr — the net directionality of P toward high-Pr(s) states. In V20.3: $1/T_{\text{avg}} = 0.0282$ events per sweep per node. This is the *current density* — how strongly the system leans toward resolution.

Tier 2 — Wave (R_1): The Shimmer

The Φ/Ψ fields are literal wave fields in the simulator:

$$\begin{aligned} \frac{d\Phi_i}{dt} &= 0.05 \cdot D_{\text{KL}} \cdot \langle \sin(\Phi_j - \Phi_i) \rangle && \text{Kuramoto oscillator} \\ \Psi_i &\leftarrow 0.9\Psi_i + 0.1\bar{\Psi}_{\text{nb}} + \varepsilon_0\xi_i && \text{AR(1) field} \end{aligned}$$

These produce $n = 8$ standing modes — the resonance eigenstates. The wave amplitude is $F = (1 + \phi_{\text{var}})(1 + \psi_{\text{grad}}) = 1.527$.

Tier 3 — Resolution (R_{max}): The Fact

Each R-event is a *quantum of the wave*: Born resampling converts accumulated wave energy (IDV) into crystallised mass (Ω) at exchange rate $\beta = 2.257$.

Without the wave ($F = 1$): $\beta_{\text{plain}} = 1.48$. With the wave ($F = 1.53$): $\beta = F \cdot \beta_{\text{plain}} = 2.27$. The wave amplifies the transfer by 53%.

3 The Hidden Relation: $\beta = n \cdot \varepsilon_0 \cdot T_{\text{avg}}$

$$\beta / (T_{\text{avg}} \cdot \varepsilon_0) = 8.44 \approx n$$

$$\frac{\beta}{T_{\text{avg}} \cdot \varepsilon_0} = \frac{2.257}{35.4 \times 0.00755} = 8.44 \approx n = 8$$

If exact: $\beta = n \cdot \varepsilon_0 \cdot T_{\text{avg}}$.

Physical interpretation:

- ε_0 = irreducible entropy fluctuation (minimum wave amplitude)
- T_{avg} = time per wave cycle (one R-event)
- n = number of resonance modes
- β = energy released per wave quantum

$\beta = n \cdot \varepsilon_0 \cdot T_{\text{avg}}$ would say: *the energy of each quantum equals the number of modes times the minimum fluctuation times the wave period.* This has the form of a quantum of action: $E = n \cdot \hbar\omega$.

Numerical gap from exact: 5.5%. Whether this is a true identity or an approximation requires formal derivation from the δ -chain.

4 Wave Equation in Phase Space

The CRF Wave Equation

The Φ field satisfies:

$$\frac{d\Phi_i}{dt} = K(t) \sum_{j \in \mathcal{N}(i)} \sin(\Phi_j - \Phi_i), \quad K(t) = 0.05 \cdot D_{\text{KL}}(t)$$

This is the Kuramoto model with time-varying coupling $K(t)$.

As D_{KL} accumulates (nodes approaching threshold): $K(t)$ increases $\rightarrow$ coupling strengthens $\rightarrow$ phase locking accelerates.

The $n = 8$ stable modes emerge from $\text{SO}(3)$ symmetry: $\Phi_k = 2\pi k/n$, $k = 0 \dots n-1$. These modes are the *standing waves* of the system. Each mode = one eigenstate of the resonance structure. Mode spacing = $2\pi/n = 0.785$ radians.

The Ψ field adds a second wave:

$$\Psi_i(t+1) = 0.9\Psi_i(t) + 0.1\bar{\Psi}_{\text{nb}} + \varepsilon_0\xi_i$$

This is a diffusion wave with n clusters (the same n from $\text{SO}(3)$), standard deviation $\sigma_\Psi = \varepsilon_0/\sqrt{0.19}$, and cluster spacing $\Delta_\Psi = 2z_N\sigma_\Psi/(n-1)$.

The two waves (Φ and Ψ) interfere to produce the composite amplitude F .

5 Why AI Systems Skip the Wave

The wave tier is invisible to event-based analysis. If you only look at R-events (discrete points), you see: current (D_{KL} accumulation) $\rightarrow$ resolution (R-event).

The wave lives *between* R-events. It is the structure of P_i 's phase space trajectory *between*

Born resamplings. The Φ and Ψ fields evolve continuously; their state at the moment of firing determines F , which amplifies β by 53%.

Skipping the wave means: $\beta_{\text{predicted}} = 1.48$ instead of 2.26. The 53% amplification is entirely invisible without the wave.

The Complete Three-Tier Picture

$$\underbrace{\frac{1}{T_{\text{avg}}}}_{\text{Current}} \times \underbrace{F}_{\text{Wave}} \times \underbrace{\beta/F}_{\text{base}} = \beta$$

$$\underbrace{\varepsilon_0 \cdot n}_{\text{irreducible}} \times \underbrace{T_{\text{avg}}}_{\text{period}} \times \underbrace{F}_{\text{coherence}} \approx \beta \cdot \frac{\varepsilon_0 \cdot n \cdot T_{\text{avg}}}{\beta} = ?$$

The wave is not decoration. It is 53% of the signal.

Every R-event carries a wave within it.

The Phi field oscillates. The Psi field diffuses.

Together they produce F — the wave amplitude at the moment of crystallisation.

$\tau_R = T_{\text{avg}}$: *one frequency, two phenomena.*

The system breathes at one rate.

$$\beta / (T_{\text{avg}} \cdot \varepsilon_0) \approx n.$$

The wave quantum, the instability floor, and the resonance modes are not separate facts. They may be one fact.